

GENERAL MEASUREMENTS:

This involves a lot of concepts & pieces to come together:

PVM - projective measurements & their statistics (done)

POVM - general measurement statistics. (last week intro)

{ Naimark Dilation } - embedding subsystem in larger closed system
Stinespring Dilation

Kraus Operators - how states update after measurement + quantum channels

Choi Matrix - channel state duality $\rightarrow$ UNIFIES THINGS.
 + semi-definite programming (SDP) for optimal measurements/coding/compression.

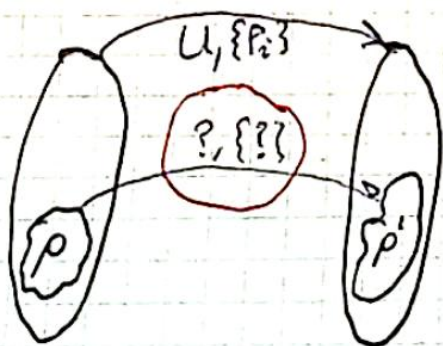
$\Rightarrow$ this is going to take a few weeks!

BASIC IDEA:

- all closed systems evolve by a unitary and measurements are projections, i.e., $U, \{P_i\}_{i=1}^n$

BUT:

- we often only see & interact with a part of a larger closed system, and restricted to that part, evolution and measurements do not look unitary & projective.

QUESTION:

- ① How do we calculate the $?, \{??\}$ directly from the nice clean larger closed system?
- ② How do we go the other way? $\left(\begin{matrix} ?, \{??\} \\ \rightarrow U, \{P_i\} \end{matrix} \right)$

- note that if a system is described by unitary evolution & projective measurements we can implement it physically. e.g., 1 qubit via particle spin $\frac{1}{2}$, photon polarization. [2]

GOAL:

We will show that systems that seem to evolve non-unitarily & do not use projective measurements can be considered subsystems of a larger closed system that does.

→ this is Naimark dilation (expansion).

→ allows use to physically realize such systems.

Recall Previous Non-Projective Examples:

① Noisy PVM.

• let $\{P_i\}$ be proj. meas'ts on a system X .

→ prob. of outcome 'j' given state ρ is:

$$p(j) = \text{tr}(P_j \rho).$$

• recording device is noisy and records 'a' given quantum output 'j' c prob'y: $q(a|j)$.

→ then
$$p(a) = \sum_j p(j) \cdot q(a|j)$$

$$= \text{tr} \left[\sum_j q(a|j) P_j \right] \rho$$

$$p(a) \equiv \text{tr} E_a \rho$$

where

$$E_a \equiv \sum_j q(a|j) P_j$$

① $E_a \geq 0$

③ E_a not projections.

② $\sum_a E_a = I$

② Trine Example:

- have an alphabet of three qubits:

$$|\psi_k\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i k/3} |1\rangle), \quad k=0,1,2.$$

i) Standard projective measurements in $\mathbb{C}^2$

$$P_{\text{succ}}^{\text{PVM}} \approx 0.622 \quad \left(\frac{2+\sqrt{3}}{6} \right)$$

ii) Use effects operators instead:

$$E_k = \frac{2}{3} |\psi_k\rangle\langle\psi_k|, \quad k=0,1,2.$$

Same: a) $E_k \geq 0$

$$b) \sum E_k = I$$

c) E_k not orthogonal projections. ($E_k^2 \neq E_k$).

$$P_{\text{succ}}^{\text{PVM}} = \frac{2}{3} \approx 0.667 > P_{\text{succ}}^{\text{PVM}}.$$

QUESTION

Nice, but how do we implement these E_k "measurements"?

- they are not projections & do not correspond to a Hermitian operator we can measure.

- only i) positive, and ii) sum to identity.

Ans: ① Go to a higher dimension by appending a system we can measure in.

② Do a unitary evolution in the larger space.

③ Do a projective measurement in the larger space.

→ we are back to unitaries & projections then AND can implement!

- we've seen this "going to a higher dimension" trick in simplifying a system with purification.

- say we have a mixed state ρ_x on the Hilbert space H_x .

$$\rho_x = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

- purify it by appending a second H.S. H_y of same r as its $\dim$. $\rightarrow H_y \otimes H_x$.

- form the pure state:

$$|\phi\rangle_{yx} = \sum_i \underset{\substack{\uparrow \\ H_y}}{|i\rangle_y} \otimes \underset{\substack{\uparrow \\ H_x}}{\sqrt{p_i} |\psi_i\rangle_x} \in H_y \otimes H_x$$

- we showed that tracing out H_y gives us ρ_x again:

$$\text{tr}_y |\phi\rangle\langle\phi|_{yx} = \rho_x$$

$\Rightarrow$ so the messy mixed state in H_x is a nice clean pure state in $H_y \otimes H_x$ by this construction.

$\rightarrow$ NOTE:

- in this case we prepend the ancilla space.

$$H_x \rightarrow H_y \otimes H_x$$

- this is convenient for picturing operations as matrices in block form later.

- recall Kronecker product of matrices

$$A \otimes B = \begin{bmatrix} a_{11}B & a_{12}B & \dots \\ a_{21}B & a_{22}B & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} \rightarrow \text{in blocks of } \dim(B).$$

Naimark Dilation Th^m: (POVM $\Rightarrow$ PVM on a larger space). 5

Let X be the system measured.

A POVM on X is a collection of effects operators, $\{E_k\}$, s.t.

① $E_k \geq 0$ (semi-positive def.), $k=1..m$

② $\sum_{k=1}^m E_k = I_X$ (completeness)...

Then there exists:

① An ancilla space Y with an orthonormal basis:
 $\{|k\rangle_Y\}_{k=1}^m$ (measurement outcome "pointers").

② A projective measurement PVM on Y :
 $\{P_k = |k\rangle\langle k|\}_{k=1}^m$

③ An isometry $V: X \rightarrow Y \otimes X$, ($V^\dagger V = I_X$).
such that for each k ,
 $E_k = V^\dagger (P_k \otimes I_X) V$

and, $p(k) = \text{tr } E_k \rho$

$$= \text{tr} \left[(P_k \otimes I_X) \underbrace{V \rho V^\dagger}_{\text{embeds } \rho \text{ in } Y \otimes X} \right]$$

• proj.'ve measurement in $Y \otimes X$.

$\Rightarrow$ NOTE: we are calculating only outcome probabilities of measurements here.

Not. state updates after measurement

$\rightarrow$ to come $\bar{c}$ Kraus operators!

Aside:

- ① Since each $E_k \geq 0$, it has semi-pos. eigenvalues ≥ 0
 $\therefore$ we can take its semi-positive squareroot via its spectral decomposition.

$$- E_k \rightarrow \sqrt{E_k} \text{ such that } (\sqrt{E_k})^2 = E_k, k=1 \dots m.$$

- ② If $V: X \rightarrow Y \otimes X$ is an isometry, it embeds X in $Y \otimes X$ and preserves distances.

$$\text{So, } \langle \psi | \psi \rangle_X = \langle (V\psi) | V\psi \rangle_{Y \otimes X}$$

$$= \langle \psi | \underbrace{V^\dagger V}_{I_X} \psi \rangle_X$$

$$\therefore V^\dagger V = I_X$$

PROOF:

- ① Let $Y \cong \mathbb{C}^m$ be an ancilla space $\vec{e}$ basis $\{|k\rangle\}_{k=1}^m$

Define projective measurements on Y by,

$$P_k = |k\rangle\langle k| \rightarrow \text{satisfy: } P_i P_j = \delta_{ij} P_i$$

$$\sum_k P_k = I_Y$$

- ② Define the isometry $V: X \rightarrow Y \otimes X$ as,

$$V = \sum_{k=1}^m |k\rangle_Y \otimes \sqrt{E_k}$$

What does V do?

Take any $|\psi\rangle \in X$. Then

$$V|\psi\rangle = \left(\sum_k |k\rangle \otimes \sqrt{E_k} \right) |\psi\rangle$$

$$\text{So, } V|\psi\rangle = \left(\sum_k |k\rangle_Y \otimes \sqrt{E_k} \right) |\psi\rangle$$

$$= \left(\sum_k |k\rangle_Y \otimes \sqrt{E_k} \right) (1_c \otimes |\psi\rangle)$$

$$= \sum_k \underbrace{|k\rangle_Y}_{\hat{Y}} \otimes \underbrace{\sqrt{E_k}}_{\hat{X}} |\psi\rangle.$$

→ this is a split into branches:

- think of $\sum_k |k\rangle \otimes \sqrt{E_k} |\psi\rangle$

$$= \sum_k \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \end{pmatrix} \otimes \sqrt{E_k} |\psi\rangle$$

← k'th
p.c.s

$$= \begin{bmatrix} \sqrt{E_1} |\psi\rangle \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \sqrt{E_2} |\psi\rangle \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \dots + \begin{bmatrix} 0 \\ \vdots \\ \sqrt{E_m} |\psi\rangle \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{E_1} |\psi\rangle \\ \sqrt{E_2} |\psi\rangle \\ \vdots \\ \sqrt{E_m} |\psi\rangle \end{bmatrix} \in Y \otimes X$$

$$= \begin{bmatrix} \sqrt{E_1} \\ \sqrt{E_2} \\ \vdots \\ \sqrt{E_m} \end{bmatrix} [|\psi\rangle] = V|\psi\rangle$$

↑ m-blocks, each block mapping $X \rightarrow X$.

= V.

③ Check the isometry of V : $V^\dagger V = I_x$.

$$\begin{aligned}
 V^\dagger V &= \left(\sum_k |k\rangle \otimes \sqrt{E_k} \right)^\dagger \left(\sum_j |j\rangle \otimes \sqrt{E_j} \right) \\
 &= \sum_k \langle k| \otimes \sqrt{E_k}^\dagger \cdot \sum_j |j\rangle \otimes \sqrt{E_j} \\
 &= \sum_{kj} \underbrace{\langle k|j\rangle}_\delta_{kj} \otimes \sqrt{E_k}^\dagger \sqrt{E_j} \\
 &= \sum_k 1 \otimes E_k \\
 &= \sum_k E_k
 \end{aligned}$$

$V^\dagger V = I_x$ by requirement of effect operators. ✓

④ Show embedding identity of effects: $E_k = V^\dagger (P_k \otimes I_x) V$.

- the idea is that $P_k \otimes I_x$ "selects the k -branch"

$$(P_k \otimes I) V = (|k\rangle\langle k| \otimes I) \left(\sum_j |j\rangle \otimes \sqrt{E_j} \right)$$

$$= \sum_j \underbrace{|k\rangle\langle k|j\rangle}_\delta_{kj} \otimes \sqrt{E_j}$$

$$= |k\rangle \otimes \sqrt{E_k}$$

$$\approx \begin{bmatrix} 0 \\ \vdots \\ \sqrt{E_k} \\ \vdots \\ 0 \end{bmatrix}$$

$$\text{So, } V^\dagger (P_k \otimes I) V$$

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$$= \left(\sum_i |i\rangle \otimes \sqrt{E_i} \right)^\dagger (|k\rangle \otimes \sqrt{E_k})$$

$$= \sum_i \underbrace{\langle i|k \rangle}_{\delta_{ik}} \otimes \sqrt{E_i}^\dagger \sqrt{E_k}$$

$$= E_k$$

$$\therefore \boxed{V^\dagger (P_k \otimes I) V = E_k} \quad (*)$$

⑤ Embed ρ in $Y \otimes X$:

- let ρ be a density operator on X .
- we embed ρ in $L(Y \otimes X)$ by:

$$\rho \rightarrow \sigma \equiv V \rho V^\dagger$$

$$\begin{aligned} \text{• note } V \rho V^\dagger &= V \cdot \sum_i p_i |\psi_i\rangle \langle \psi_i| \cdot V^\dagger \\ &= \sum_i p_i \underbrace{V |\psi_i\rangle \langle \psi_i| V^\dagger}_{Y \otimes X} \in L(Y \otimes X). \end{aligned}$$

⑥ Now measure $P_k \otimes I$ on the embedded σ :

$$\begin{aligned} p_{\text{emb}}(k) &= \text{tr} (P_k \otimes I) \sigma \\ &= \text{tr} (P_k \otimes I) V \rho V^\dagger \\ &= \text{tr} V^\dagger (P_k \otimes I) V \rho \\ &= \text{tr} E_k \rho \quad \text{from } (*) \text{ above.} \\ &= p(k). \end{aligned}$$

• the measurement statistics match by construction!

⑦ Where's the unitary?

- we create it from the isometry $V: X \rightarrow Y \otimes X$ by extending it to a unitary $U: Y \otimes X \rightarrow Y \otimes X$.

- write $U = \begin{bmatrix} V \\ U' \end{bmatrix}$ where $V \sim (\dim Y \otimes X) \times (\dim X)$.
 $U \sim (\dim Y \otimes X) \times (\dim Y \otimes X)$.
 fill in the U' to make U unitary.
 ↑ first block column.

- note we can write $V|\psi\rangle = U(|0\rangle \otimes |\psi\rangle)$

$$\begin{aligned}
 &= \begin{bmatrix} V \\ \vdots \end{bmatrix} \begin{bmatrix} |\psi\rangle \\ 0 \\ \vdots \\ 0 \end{bmatrix} \\
 &= \begin{bmatrix} \sqrt{E_1} \\ \sqrt{E_2} \\ \vdots \\ \sqrt{E_n} \end{bmatrix} \begin{bmatrix} |\psi\rangle \\ 0 \\ \vdots \\ 0 \end{bmatrix} \\
 &= \sqrt{E_1}|\psi\rangle + \sqrt{E_2}|\psi\rangle + \dots + \sqrt{E_n}|\psi\rangle \\
 &= \sum_k |k\rangle \otimes \sqrt{E_k}|\psi\rangle \quad \checkmark
 \end{aligned}$$

⑧ Any unitary (a matrix in general) can be written:

$$U = \sum_{ij} |i\rangle\langle j| \otimes B_{ij} = \begin{bmatrix} B_{11} & B_{12} & \dots \\ B_{21} & B_{22} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

where B_{ij} 's operate on X here: $B_{ij}: X \rightarrow X$.

- they are arbitrary in general but if U is unitary we require: $U^\dagger U = U U^\dagger = I_{Y \otimes X}$.

Check what conditions on the B_{ij} the unitary $U^\dagger U = I$ forces: II

$$\begin{aligned}
 U^\dagger U &= \left(\sum_{ij} |j\rangle\langle i| \otimes B_{ij}^\dagger \right) \left(\sum_{kl} |k\rangle\langle l| \otimes B_{kl} \right) \\
 &= \sum_{\substack{ij \\ kl}} |j\rangle\langle i| |k\rangle\langle l| \otimes B_{ij}^\dagger B_{kl} \\
 &\quad \underbrace{\delta_{ik} \rightarrow \text{forces } k=i}_{\text{}} \\
 &= \sum_{ij} |j\rangle\langle i| \otimes B_{ij}^\dagger B_{ii}
 \end{aligned}$$

For this to equal $I_{Y \otimes X}$ then each of the two sums equal I_Y & I_X respectively.

$$I_Y \otimes I_X = \underbrace{\sum_{jl} |j\rangle\langle l|}_{\textcircled{A}} \otimes \underbrace{\sum_i B_{ij}^\dagger B_{ii}}_{\textcircled{B}}$$

$$\textcircled{A} = I_Y \rightarrow \sum_{jl} |j\rangle\langle l| = I_Y$$

$\textcircled{B}$ To force $\textcircled{A} = I_Y$ we must have a δ_{jl} factor in $\textcircled{B}$.
 $\therefore \sum_i B_{ij}^\dagger B_{il} = \delta_{jl} I_X \rightarrow \text{columns as blocks are orthog.}$

$$\text{So, } \boxed{\sum_i B_{ik}^\dagger B_{il} = \delta_{kl} I_X} \rightarrow \text{same column (k'th) sum to identity.}$$

- **NOTE:** Similar result for rows from $U^\dagger U = I$.

⑨ What does this mean for our U ?

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$$U = \begin{bmatrix} V & U' \end{bmatrix} = \begin{bmatrix} \sqrt{E_1} & & \\ & \sqrt{E_2} & \\ & & \ddots \\ & & & \sqrt{E_m} \end{bmatrix} \begin{bmatrix} & U' \end{bmatrix}$$

This gives us for the first column:

$$\sum_k \sqrt{E_k}^\dagger \sqrt{E_k} = I_x$$

OR $\boxed{\sum_k E_k = I_x}$ ← completeness condition on the effects, $\{E_k\}$!

⑩ Putting it all Together:

We can write: $V|\psi\rangle = U(|0\rangle \otimes |\psi\rangle)$

$$= U(|0\rangle \otimes I_x)(1 \otimes |\psi\rangle)$$

$$V|\psi\rangle = U(|0\rangle \otimes I_x)|\psi\rangle$$

$$\Rightarrow V = U(|0\rangle \otimes I_x)$$

$$E_k = (\langle 0| \otimes I_x) U^\dagger (P_k \otimes I_x) U (|0\rangle \otimes I_x)$$

↑ projective measurement in large space
↑ unitary
↑ Embed in larger dim^s space

- ⑪ This is the:
- Embed in higher dim^s space
 - Unitary transformation
 - Projective measurement.